

# TOHEEB SALAUDEEN

Full-Stack Software Engineer · International Relations · Obafemi Awolowo University

## CONTACT

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## LINKS

[Portfolio](#) —  
[roiwhiz.vercel.app](https://roiwhiz.vercel.app)  
[GitHub](#) —  
[github.com/Roiwhiz](https://github.com/Roiwhiz)  
[LinkedIn Profile](#)  
[GeoWatch Demo](#)

## EDUCATION

**B.Sc. International Relations**  
Obafemi Awolowo  
University, Nigeria  
2023 – 2027  
Year 3 of 4

## FRONTEND

HTML · CSS · JavaScript  
TypeScript · React  
Next.js · Tailwind CSS  
Zustand · React Query  
next-intl · shadcn/ui  
Motion (animations)

## BACKEND

Node.js · Express.js  
TypeScript · Prisma  
PostgreSQL · Redis  
REST API Design  
Session Management  
Error Architecture

## AI & INTEGRATIONS

Gemini API · OpenAI API  
Tavily Search API  
ACLED Conflict API  
UN Comtrade API  
World Bank API  
IMF Data API  
RAG · Vector Search  
Agent Loop Design

## TOOLS & PLATFORMS

Git · GitHub · VS Code

## PROFESSIONAL SUMMARY

Self-taught full-stack engineer who went from a first online class in mid-2024 to building and deploying production-grade AI systems — with no formal CS background and a full university course load. I came to software from International Relations and Digital Marketing, which means I understand the problems I am building for, not just the code that runs them. I design before I build, write typed and structured systems, and ship things that actually work.

## PROJECTS

**GeoWatch — Autonomous Geopolitical Intelligence Platform** 2025 – Present  
*Next.js · Express · TypeScript · PostgreSQL · Redis · Prisma · Gemini API · Tavily · Deployed*

- **Agent loop:** custom-built on Gemini 2.5 Flash, orchestrating live web search, ACLED conflict data, and UN Comtrade trade flows to produce structured intelligence reports with IR framework classification (Realism, Liberalism, Constructivism, Political Economy).
- **Session architecture:** Redis-based locking prevents concurrent agent runs; full multi-turn conversation history persists in PostgreSQL and is hydrated on every agent call.
- **Error handling:** typed error class hierarchy across the full stack — Gemini, Prisma, and application errors classified and handled by a global Express middleware that never exposes internals in production.
- **Response classification:** classifier detects structured IR reports vs conversational replies and routes each to a different frontend renderer.
- **Internationalisation:** fully localised in 6 languages (EN, FR, ES, DE, AR, ZH) with RTL layout support for Arabic.

**Conflict Data Dashboard** *In Development*  
*Next.js · TypeScript · ACLED API · Mapbox · Recharts*

- Interactive conflict intelligence dashboard pulling live from the ACLED API — filterable map of conflict events by country, actor, and event type with fatality trend lines and actor breakdowns.

**Trade Flow Visualiser** *In Development*  
*Next.js · TypeScript · UN Comtrade API · D3.js*

- Takes any two countries and a year, queries UN Comtrade, and renders bilateral trade as a Sankey diagram — showing what each exports to the other, in what volumes, and how that has shifted over time.

**Country Risk Intelligence Brief Generator** *In Development*  
*Next.js · TypeScript · Gemini API · World Bank API · IMF API*

- AI-powered risk analysis that pulls live World Bank and IMF data, runs targeted web searches, and synthesises everything into a structured political and economic risk brief — built on the same agent architecture as GeoWatch.

**UN Resolutions Search Engine** *In Development*  
*Next.js · TypeScript · PostgreSQL · pgvector · OpenAI Embeddings*

- Semantic search engine over UN Security Council resolutions using RAG — embeds the full resolution corpus into a vector database and returns ranked

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| <div>Postman · Prisma Studio</div> <div>Vercel · Render</div> <div>pgvector · Mapbox</div> <div>CORE STRENGTHS</div> <div>System Design First</div> <div>Full-Stack Architecture</div> <div>AI Agent Systems</div> <div>Data Visualisation</div> <div>Database Design</div> <div>Multi-language UX</div> <div>LANGUAGES</div> <div>English — Native</div> <div>Arabic (MSA)</div> <div>Professional Proficiency</div> | <div>results with AI-generated summaries for natural language queries.</div> <div><div>Sanctions Tracker</div><div><i>In Development</i></div><div><i>Next.js · TypeScript · PostgreSQL · OFAC · EU Sanctions Data</i></div><div><div>– Aggregates active sanctions from US OFAC, the EU, and the UN into a single queryable interface — filterable by country, entity, issuing body, sector, and designation date.</div></div></div> <div><div>Multilingual News Aggregator</div><div><i>In Development</i></div><div><i>Next.js · TypeScript · PostgreSQL · Gemini API · RSS</i></div><div><div>– Pulls geopolitical news from RSS feeds across multiple languages, translates and clusters articles by topic using an LLM, and surfaces divergent narratives on the same event side by side.</div></div></div> <div><div>Treaty Database</div><div><i>In Development</i></div><div><i>Next.js · TypeScript · PostgreSQL · UN Treaty Collection API</i></div><div><div>– Full-text searchable database of international treaties from the UN Treaty Collection — filterable by signatory country, topic area, status, and date range.</div></div></div> <div><div>Refugee and Displacement Tracker</div><div><i>In Development</i></div><div><i>Next.js · TypeScript · UNHCR API · Mapbox · Recharts</i></div><div><div>– Visualises forced displacement trends from UNHCR data on an interactive map with country profiles, historical trend lines, and event-based annotations — built for NGO researchers, humanitarian workers, and journalists.</div></div></div> <div><div>UN General Assembly Voting Tracker</div><div><i>In Development</i></div><div><i>Next.js · TypeScript · PostgreSQL · D3.js · UN GA Voting Records</i></div><div><div>– Ingests the full UN General Assembly voting record and models it as a network graph and heatmap — showing voting blocs, alignment clusters, and how countries' positions shift over time on issues like sovereignty, human rights, and climate.</div></div></div> <div><div>BACKGROUND</div><div><div>Self-Taught Full-Stack Developer</div><div><i>2024 – Present</i></div><div>Started with a multi-week online class organised by a residential mate in mid-2024. The class made a strong enough impression to keep going independently. Progressed from Udemy courses and YouTube through to building production systems entirely from documentation — consistently, alongside a full academic course load.</div></div><div><div>Digital Marketing</div><div><i>2023 – 2024</i></div><div>Completed Google Digital Marketing certifications and studied SEO, SEM, and content strategy independently. Financial constraints on tooling led to a pivot toward software engineering — a domain where the barrier to entry is a computer and discipline, not a subscription.</div></div></div> |
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